Studying volcanic explosions 11a

The different materials that erupt out of volcanoes can be classified into solids, liquids and gases.

Solids
Solid materials are thrown from volcanoes, ranging from huge blocks of solidified lava to fine dust particles. The volcanic solids are grouped according to their size.

Blocks	greater than 24 cm
Bomb	between 5 – 24 cm (mean 16 cm)
Lapilli	between 3 – 5 cm (mean 4 cm)
Ash	between 0.05 – 3 cm (mean 2 cm)
Dust	less than 0.05 cm

Liquids
The hot liquid material that (comes/came) out of a volcano (is called/are called) lava. Chemicals (in/on) the lava, such as silica, (affect/effect) (its/it's) viscosity. For example, lava with a high viscosity (moves/move) very (slow/slowly) and lava with a low viscosity (move/moves) very (fast/fastly).

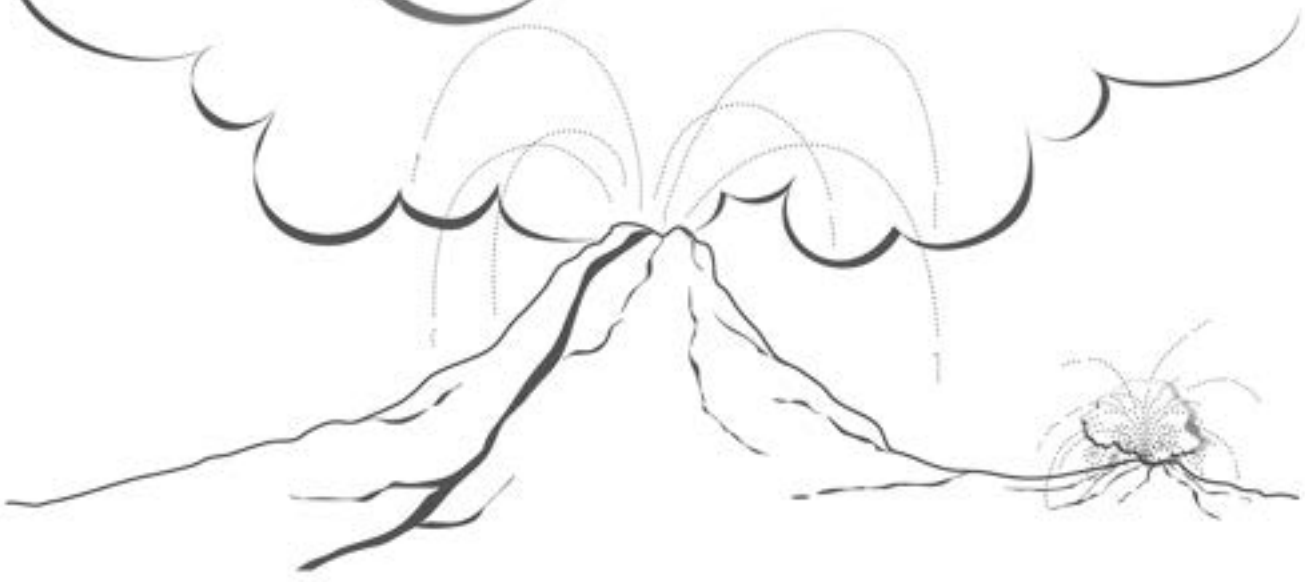
Lava that (have/has) a very high silica content (of/off) about 70–80% (is/are) very viscous. This lava (does not/does) flow very easily and (plugged/plugs) up volcanic vents. These types of lavas (are called/is called) **rhyolites**.

Lava with a high (silica/sillica) content of about 60–69% (produce/produces) **dacite** lava. This lava (flows/flow) slowly.

Andesite lava (has/have) a silica content of about 54–59% and (move/moves) much faster than dacite lava. **Basaltic** lava, (on/in) the other hand, (has/have) a relatively low silica content of about 45–58%. These lavas (is/are) the least viscous and (flow/flows) quickly for many (kilometers/kilometres) away from the volcanic (vent/went).

Gases
Volcanologists have collected samples of gas from different volcanoes and have come up with the following components of gases.

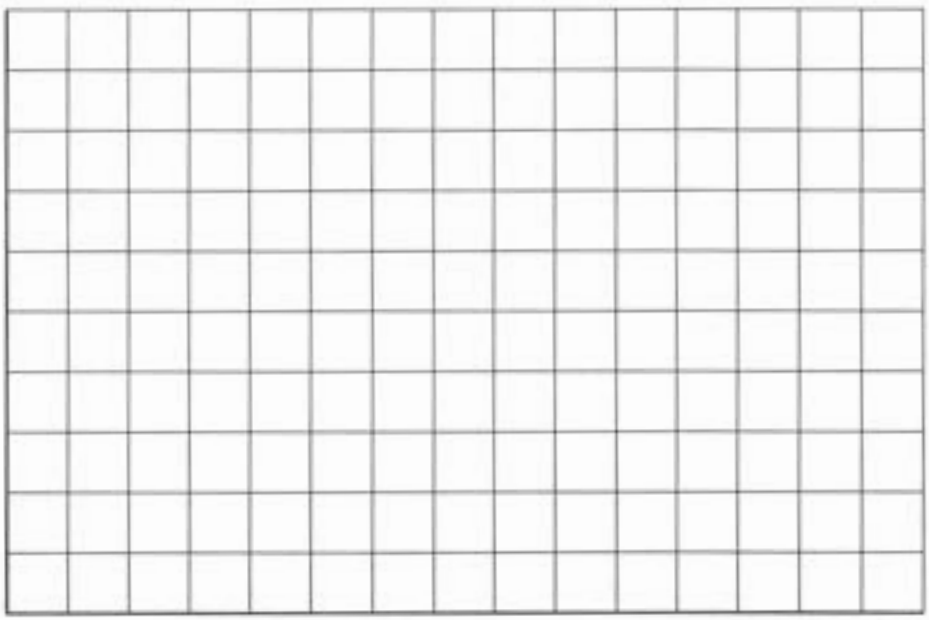
Water	**76%**
Carbon dioxide	**12%**
Sulfur dioxide	**6%**
Nitrogen	**3.4%**
Carbon monoxide	**1%**
Hydrogen	**1%**
Sulfur	**0.4%**
Chlorine	**0.1%**
Argon	**0.1%**



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Studying volcanic explosions 11b

1 Draw a column graph in the space provided below to display the composition of volcanic gas. The gas components go along the horizontal axis.



2 Cross out the incorrect terms in the text on liquids on the previous page, leaving the correct term.

3 Complete the table below about the liquids being spewed out of a volcano.

Name of lava type	Viscosity (high, medium, low)	Speed of lava flow (fast, slow...)	Silica content (%)

4 Draw a scaled diagram for each of the solids described in the text. Use the scale 1: 4 (ie, 1 scaled centimetre equals 4 real centimetres).

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